

Software Test Plan (STP)

Project: Online Internship & Placement Preparation Portal

Version: 1.0

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Team Members & Contribution

Name	SRN	Contribution
Patnaikuni Sai Dheeraj	PES2UG24CS341	Section 1 (Introduction), Section 9 (Roles & Responsibilities); overall document structuring, compilation, and final review
Nikhil Raj G	PES2UG24CS319	Sections 2-4 (Test Items, Features to be Tested, Features Not to be Tested), Section 6 (Test Environment), and functional test cases in Section 13.1
NAVYA R	PES2UG24CS308	Section 5 (Test Approach/Strategy including 5.1 Security Validation), Section 10 (Risks & Mitigation), and security/NFR test cases in Section 13.1
Sohel Mahibubasa Chapparband	PES2UG25CS826	Sections 7-8 (Test Schedule, Test Deliverables), Sections 11-12 (Assumptions & Dependencies, Suspension & Resumption Criteria), Section 13.2 (Traceability/RTM), and Section 14 (Test Metrics & Reporting)

Revision History

Version	Date	Author	Change Summary	Approval
1.0	16-09-2026	Patnaikuni Sai Dheeraj, Nikhil Raj G, NAVYA R, Sohel Mahibubasa Chapparband	Initial Test Plan drafted for Online Internship & Placement Preparation Portal	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	Sheela M Devi		
Faculty Guide			

1. Introduction

1.1 Purpose

This document defines the test plan for the Online Internship & Placement Preparation Portal v1.0. It outlines the testing objectives, scope, strategy, resources, schedule, and responsibilities required to verify that the system meets the functional, non-functional, and security requirements defined in the SRS.

1.2 Scope

Testing covers student registration and authentication, student profile management, internship discovery and application, application-status tracking, aptitude tests and scoring, test-history storage, placement-preparation progress, performance analytics, recruiter internship management, recruiter application management, and administration functions. Non-functional and security validation are also included. External third-party service internals and infrastructure outside the project boundary are excluded.

1.3 References

- Online Internship & Placement Preparation Portal — Software Requirements Specification (SRS), Version 1.0.
- Online Internship & Placement Preparation Portal — Software Architecture and Design Specification (SAD), Version 1.0.
- Provided Software Test Plan (STP) template for the Software Engineering mini-project.
- Project requirements and acceptance criteria defined in the SRS.

1.4 Definitions

- SRS — Software Requirements Specification; STP — Software Test Plan; RTM — Requirements Traceability Matrix.
- FR — Functional Requirement; NFR — Non-Functional Requirement; RBAC — Role-Based Access Control.
- API — Application Programming Interface; UAT — User Acceptance Testing.

2. Test Items

- User Registration & Authentication Module
- Student Profile Management Module
- Internship Search and Application Module
- Aptitude Test Module
- Placement Preparation Progress and Analytics Module
- Recruiter Module
- Administration Module
- REST API and frontend-backend integration
- MongoDB data persistence and integrity
- Security controls and input validation

3. Features to be Tested

The following features are mapped directly to the requirement IDs defined in the SRS.

Req ID	Feature / Requirement
IPR-F-001	Student registration
IPR-F-002	User login/authentication
IPR-F-003	Role-based access for Student, Recruiter and Admin
IPR-F-004	Create and update student profile
IPR-F-006	Display available internship opportunities
IPR-F-007	Search and filter internships
IPR-F-008	Apply for an available internship
IPR-F-009	Prevent duplicate internship applications
IPR-F-011	Take aptitude tests
IPR-F-012	Automatically calculate aptitude-test score
IPR-F-013	Store test scores and test history
IPR-F-016	Recruiter internship management
IPR-F-017	Recruiter application management
IPR-NF-001	Dashboard response time ≤ 3 seconds under normal conditions
IPR-NF-003	Support ≥ 500 concurrent users under planned deployment

IPR-NF-004	Responsive UI across desktop, tablet and mobile
IPR-SR-001	Authentication protection
IPR-SR-004	HTTPS/TLS
IPR-SR-005	Input sanitization

4. Features Not to be Tested

- Internal implementation of third-party hosting, browser, database, or external service infrastructure.
- Underlying cloud-provider infrastructure provisioning and physical network hardware.
- Internal implementation details of MongoDB or Node.js/Express.js beyond the project's application interfaces.
- External notification/email service infrastructure, if used, outside the project boundary.

5. Test Approach / Strategy

Levels

- Unit tests (module/service level, e.g. authentication, internship, aptitude-test and progress services).
- Integration tests (React frontend ↔ REST API ↔ MongoDB).
- System tests (end-to-end flows across student, recruiter and admin interfaces).
- Acceptance tests (UAT using representative internship and placement-preparation scenarios).

Types

- Functional testing (core portal features).
- Regression testing (after each build/change).
- Performance testing (dashboard response time and concurrent-user load).
- Security testing (authentication, authorization, password protection and input sanitization).
- Usability testing (UI clarity, accessibility and responsiveness).

Entry Criteria

A stable build is delivered; the test environment (frontend, backend and database) is provisioned and reachable; test accounts and representative internship/test data are available.

Exit Criteria

100% of planned test cases are executed; zero critical or high-severity defects remain open; applicable SRS acceptance criteria are satisfied; NFR and security targets are met or formally waived.

5.1 Security Validation

- Validate authentication protection and rejection of invalid/expired sessions.
- Verify passwords are stored using salted adaptive hashing and are never stored in plaintext.
- Verify role authorization prevents students, recruiters and admins from accessing unauthorized functions.
- Verify HTTPS/TLS is enforced for REST communication in the secure deployment.
- Fuzz-test form/API inputs for malformed and injection-style data.

- Verify server-side sanitization prevents stored or reflected XSS from user-controlled fields.
- Verify rate limiting or controlled rejection for repeated authentication attempts where implemented.

6. Test Environment

- Hardware: Standard developer laptops/workstations; representative mobile/tablet devices or responsive viewport emulation.
- Software: React.js frontend, Node.js + Express.js backend, MongoDB, modern Chrome/Firefox/Edge/Safari browsers.
- Tools: Postman for REST API testing; Selenium/Playwright for UI automation; JMeter/Artillery for load testing; Jira for defect tracking.
- Test Data: Dummy Student, Recruiter and Admin accounts; internship records; applications; aptitude questions/results; preparation-progress records.

7. Test Schedule

- Test case design: 18-Sep-2026
- Environment setup: 20-Sep-2026
- Test execution start: 22-Sep-2026
- Test execution end: 26-Sep-2026
- UAT: 27-Sep-2026 to 28-Sep-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead / Project Lead	Patnaikuni Sai Dheeraj	Prepare test plan, coordinate testing, oversee security validation, traceability and final review/submission.
Test Engineer	Nikhil Raj G	Design and execute functional/integration test cases, record defects and support re-testing.
Test Engineer	NAVYA R	Design and execute security/NFR test cases, support risk analysis and security validation.

Developer / Test Support	Sohel Mahibubasa Chapparband	Support test environment/data, maintain RTM and assist defect fixes, triage and re-testing.
Developer	All team members	Support defect fixes, integration and re-testing.

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke-test builds and prioritize testing of core modules.
Test environment downtime	Maintain a reproducible local test environment and database backup/seed data.
Incomplete or inconsistent test data	Prepare seeded Student, Recruiter, Admin, internship, application and aptitude-test data before execution.
API/frontend integration failures	Use Postman and integration checks early; maintain clear API contracts.
Performance issues under concurrent load	Use an isolated load-test environment and monitor response time/resource usage.

11. Assumptions & Dependencies

- The portal build and core modules will be stable and available for the scheduled test window.
- Test data and role-specific accounts will be provided or seeded before execution begins.
- MongoDB and backend services will be provisioned and reachable throughout testing.
- Frontend and backend API contracts will remain sufficiently stable during the planned test cycle.

12. Suspension & Resumption Criteria

Suspend testing if: the test environment is unavailable for more than 4 hours, or the build is unstable enough to block more than 30% of planned test cases.

Resume testing if: blocking defects are resolved and the environment/build is confirmed stable.

13. Test Case Management & Traceability

13.1 Test Cases

A minimum set of planned test cases is defined below, covering authentication, profile, internship, aptitude-test, performance, usability, and security requirements.

TC ID	Req ID	Description	Preconditions	Steps	Expected Result	Priority
TC-Auth-01	IPR-F-001	Register with valid unique student details	Registration page/API reachable	Submit valid unique personal and academic details	Student account is created and success response is returned	High
TC-Auth-02	IPR-F-002	Login with valid credentials	Registered account exists	Enter correct email/password and submit	User is authenticated and dashboard is displayed	High
TC-Auth-03	IPR-F-003	Role-based access control	Student, Recruiter and Admin accounts exist	Attempt to access role-specific pages/APIs using each role	Only authorized functions are accessible; unauthorized requests are rejected	High
TC-Prof-01	IPR-F-004	Create and update student profile	Authenticated student account exists	Enter profile details, save, then edit and save again	Profile is created and updated values persist correctly	High

TC-Int-01	IPR-F-006	Display available internship opportunities	Internship records exist in database	Open internship listing as a student	Available internships are displayed with relevant details	High
TC-Int-02	IPR-F-007	Search and filter internships	Multiple internship records exist	Search by keyword and apply available filters	Only matching internships are displayed	High
TC-Int-03	IPR-F-008	Apply for an available internship	Authenticated student and open internship exist	Submit an application with required details	Application is created and confirmation/status is shown	High
TC-Int-04	IPR-F-009	Prevent duplicate application	Student has already applied to the internship	Submit another application for the same internship	Duplicate application is rejected and original application remains intact	High
TC-Test-01	IPR-F-011	Attempt an available aptitude test	Authenticated student and available test exist	Open test, answer questions and submit	Test is completed and submission is accepted	High
TC-Test-02	IPR-F-012	Automatic aptitude-test score calculation	Completed test with known correct answers	Submit test and compare displayed score with expected score	System calculates and displays the correct score automatically	High
TC-Perf-01	IPR-NF-001	Dashboard response time under normal load	Test environment configured	Open dashboard repeatedly under normal simulated usage	Dashboard response time is ≤ 3 seconds	High
TC-Scale-01	IPR-NF-003	Concurrent user capacity	Load-testing tool configured	Ramp test load to 500 concurrent users	System remains stable and supports the planned concurrent load	High
TC-UI-01	IPR-NF-004	Responsive UI across devices	Application deployed to test environment	Open key pages at desktop, tablet and mobile viewport sizes	Layout remains usable and key controls remain accessible	Medium
TC-Sec-01	IPR-SR-001	Authentication protection	Valid and invalid credentials available	Attempt valid login, invalid login and access protected page without login	Valid users authenticate; invalid/unauthenticated access is rejected	High
TC-Sec-04	IPR-SR-004	HTTPS/TLS enforcement	Secure deployment/test server available	Attempt HTTP access and inspect secure HTTPS access	Secure HTTPS/TLS connection is enforced; insecure access is rejected or redirected	High
TC-Sec-05	IPR-SR-005	Input sanitization	Authenticated form/API available	Submit malicious or malformed input such as script payloads	Input is sanitized/rejected and unsafe content is not executed or persisted	High

13.2 Requirements Traceability Matrix (RTM)

The RTM connects selected SRS requirements to executable test cases and provides the basis for coverage tracking.

Req ID	Requirement	Test Case(s)
IPR-F-001	Student registration	TC-Auth-01
IPR-F-002	User login/authentication	TC-Auth-02
IPR-F-003	Role-based access	TC-Auth-03
IPR-F-004	Student profile management	TC-Prof-01
IPR-F-006	Internship listings	TC-Int-01
IPR-F-007	Search/filter internships	TC-Int-02
IPR-F-008	Apply for internship	TC-Int-03
IPR-F-009	Prevent duplicate application	TC-Int-04
IPR-F-011	Take aptitude test	TC-Test-01
IPR-F-012	Automatic score calculation	TC-Test-02
IPR-NF-001	Dashboard response time ≤ 3 s	TC-Perf-01
IPR-NF-003	≥ 500 concurrent users	TC-Scale-01
IPR-NF-004	Responsive desktop/tablet/mobile UI	TC-UI-01
IPR-SR-001	Authentication protection	TC-Sec-01
IPR-SR-004	HTTPS/TLS	TC-Sec-04
IPR-SR-005	Input sanitization	TC-Sec-05

14. Test Metrics & Reporting

Metrics collected: % test cases executed, % passed/failed, defect density, defect aging, requirement coverage.

Reports: Daily execution status report; final Test Summary Report at the end of the execution cycle.

15. Approvals

Role	Name	Signature / Date
QA Lead / Project Lead	Patnaikuni Sai Dheeraj	Dheeraj / 16/09/2026
Dev Lead	Sohel	Sohel / 16/09/2026
Faculty / Product Owner		